

Taylor and Mclaurin Polynomial

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Taylor series

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(x_0) (x-x_0)^k}{k!}$$

derivative.

- * $x_0 = 0$ for Maclaurin
- * Difference between series & polynomial
- * $f'''(x) = f^3(x)$

$$= \frac{f^{(0)}(x_0) (x-x_0)^0}{0!} + \frac{f'(x_0) (x-x_0)^1}{1!} + \frac{f''(x_0) (x-x_0)^2}{2!} + \dots$$

$$\sum_{k=0}^5 \frac{f^{(k)}(1) (x-1)^k}{k!} = \frac{f^{(0)}(1) (x-1)^0}{0!} + \frac{f'(1) (x-1)^1}{1!} + \frac{f''(1) (x-1)^2}{2!} + \frac{f'''(1) (x-1)^3}{3!} + \frac{f^{(4)}(1) (x-1)^4}{4!} + \frac{f^{(5)}(1) (x-1)^5}{5!}$$

Taylor series of $\cos x$ at $x = \pi/2$

$$f(x) = \cos x$$

$$f'(x) = -\sin x$$

$$f''(x) = -\cos x$$

$$f'''(x) = \sin x$$

$$f^{(4)}(x) = \cos x$$

$$f(\pi/2) = \cos \pi/2 = 0$$

$$f'(\pi/2) = -\sin \pi/2 = -1$$

$$f''(\pi/2) = -\cos \pi/2 = 0$$

$$f'''(\pi/2) = \sin \pi/2 = 1$$

$$f^{(4)}(\pi/2) = \cos \pi/2 = 0$$

$$\sum_{k=0}^{+\infty} \frac{f^{(k)}(\pi/2) (x - \pi/2)^k}{k!}$$

$$= \frac{f(\pi/2) (x - \pi/2)^0}{0!} + \frac{f'(\pi/2) (x - \pi/2)^1}{1!}$$

$$+ \frac{f''(\pi/2) (x - \pi/2)^2}{2!} + \frac{f'''(\pi/2) (x - \pi/2)^3}{3!}$$

$$+ \frac{f^{(4)}(\pi/2) (x - \pi/2)^4}{4!} + \frac{f^{(5)}(\pi/2) (x - \pi/2)^5}{5!}$$

$$= 0 + (-1)(x - \pi/2) + 0 + \frac{1(x - \pi/2)^3}{3!} + 0 + \frac{(-1)(x - \pi/2)^5}{5!} + \dots$$

$$= -(x - \pi/2) + \frac{(x - \pi/2)^3}{3!} - \frac{(x - \pi/2)^5}{5!} + \dots$$

$$n\text{th term} = \frac{(-1)^{k+1} (x - \pi/2)^{2k+1}}{(2k+1)!}$$

Intuition

$(2k+1) \rightarrow$ odd series

$2k \rightarrow$ even series.

Taylor Polynomial

$$P_0(x) = f(x_0)$$

$$P_1(x) = f(x_0) + f'(x_0)(x - x_0)$$

$$P_2(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)(x - x_0)^2}{2!}$$

Maclaurin Polynomial $\Rightarrow \boxed{x_0 = 0}$

$$P_0(x) = f(0)$$

$$P_1(x) = f(0) + f'(0)x$$

$$P_2(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2!}$$

$$P_3(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \frac{f'''(0)x^3}{3!} \rightarrow$$

Expand $y = \ln x$ in the power of $x-2$

$$f(x) = \ln x$$

$$f'(x) = \frac{1}{x} = x^{-1}$$

$$f''(x) = -1x^{-2}$$

$$f'''(x) = (-1)(-2)x^{-3}$$

$$f^{(4)}(x) = (-1)(-2)(-3)x^{-4}$$

$$f(2) = \ln 2$$

$$f'(2) = 2^{-1} = 1/2$$

$$f''(2) = -(2)^{-2} = -1/4$$

$$f'''(2) = 2(2)^{-3} = 1/4$$

$$f^{(4)}(2) = -6(2)^{-4} = -\frac{6}{16} = -\frac{3}{8}$$

$$\sum_{k=0}^{+\infty} \frac{f^{(k)}(2)(x-2)^k}{k!} = f(2) + \frac{f'(2)(x-2)^1}{1!} + \frac{f''(2)(x-2)^2}{2!} + \frac{f'''(2)(x-2)^3}{3!} + \dots$$

$$= \ln 2 + 1/2(x-2) + \frac{(-1/4)(x-2)^2}{2!} + \frac{1/4(x-2)^3}{3!} + \dots$$

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Find the Maclaurin polynomial P_0, P_1, P_2, P_3 for $e^x \cos x$

$$f(x) = e^x \cos x$$

$$f'(x) = e^x \cos x - e^x \sin x$$

$$f''(x) = e^x \cos x - e^x \sin x - e^x \sin x - e^x \cos x \\ = -2e^x \sin x$$

$$f'''(x) = -2e^x \sin x - 2e^x \cos x$$

$$f(0) = e^0 \cos 0 = 1$$

$$f'(0) = 1 - 0 = 1$$

$$f''(0) = -2 \times 0 = 0$$

$$f'''(0) = 0 - 2 = -2$$

$$P_0(x) = f(0) = 1$$

$$P_1(x) = f(0) + f'(0)x = 1 + 1 \times x = 1 + x$$

$$P_2(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2!} = 1 + 1 \times x + \frac{0 \times x^2}{2!} \\ = 1 + x$$

$$P_3(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \frac{f'''(0)x^3}{3!}$$

$$= 1 + x + \frac{(-2)x^3}{3!}$$

$$= 1 + x - \frac{1}{3}x^3$$

Find the Maclaurin series for $\ln(x+1)$ and use this result to find the Maclaurin series of $\ln(1-4x^2)$.

Maclaurin series of $\ln(x+1)$:

$$\ln(x+1) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots + (-1)^{n-1} \frac{x^n}{n} + \dots \quad (1)$$

$$\ln(1-4x^2)$$

$$= \ln(1^2 - (2x)^2)$$

$$= \ln(1+2x)(1-2x)$$

$$= \ln(1+2x) + \ln(1-2x) \quad (\text{logarithmic operation})$$

Put $x=2x$ in $\ln(x+1)$ expression (1)

$$\ln(1+2x) = 2x - \frac{(2x)^2}{2} + \frac{(2x)^3}{3} - \frac{(2x)^4}{4} + \dots$$

$$= 2x - 2x^2 + \frac{8}{3}x^3 - 4x^4 + \dots$$

Put $x=-2x$ in $\ln(x+1)$ expression (1)

$$\ln(1-2x) = (-2x) - \frac{(-2x)^2}{2} + \frac{(-2x)^3}{3} - \frac{(-2x)^4}{4} + \dots$$

$$= -2x - 2x^2 - \frac{8}{3}x^3 - 4x^4 + \dots$$

$$\therefore \ln(1-4x^2) = \ln(1+2x) + \ln(1-2x)$$

$$= (2x - 2x^2 + \frac{8}{3}x^3 - 4x^4 + \dots) + (-2x - 2x^2 - \frac{8}{3}x^3 - 4x^4 + \dots)$$

$$= -4x^2 - 8x^4 + \dots$$

Find the Maclaurin series for $f(x) = (1+x)^k$ where k is any real number. If k is zero, what will happen with this series?

Maclaurin series for $f(x) = (1+x)^k$

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = f(0) + f'(0)x + \frac{f''(0)}{2!} x^2 + \frac{f'''(0)}{3!} x^3 + \dots$$

$$f(x) = (1+x)^k$$

$$f'(x) = k(1+x)^{k-1}$$

$$f''(x) = k(k-1)(1+x)^{k-2}$$

$$f'''(x) = k(k-1)(k-2)(1+x)^{k-3}$$

$$f(0) = 1$$

$$f'(0) = k$$

$$f''(0) = k(k-1)$$

$$f'''(0) = k(k-1)(k-2)$$

$$f(x) = (1+x)^k$$

$$= 1 + kx + \frac{k(k-1)}{2!} x^2 + \frac{k(k-1)(k-2)}{3!} x^3 + \dots$$

for $k \geq 0$, the series becomes :-

$$= 1 + 0 \times x + \frac{0(0-1)}{2!} x^2 + \frac{0(0-1)(0-2)}{3!} x^3 + \dots$$

$$= 1$$

$$f(x) = (1+x)^0 = 1$$

So, now, there is only one ^{constant} term. It is no more a series.

Additional information:

If k is a real number, then the Maclaurin series for $(1+x)^k$ is basically a binomial series

Write down $P_2(x)$, $P_3(x)$ and $P_4(x)$
for the Taylor series of $f(x) = \exp(-6x)$
about $x = -4$. Graph the function and any
one of the Taylor polynomials on the same
graph for the interval $[-8, -2]$.

$$f(x) = e^{-6x}$$

$$f(x) = e^{-6x}$$

$$f'(x) = -6e^{-6x}$$

$$f''(x) = (-6)^2 e^{-6x} = 36e^{-6x}$$

$$f'''(x) = (-6)^3 e^{-6x} = -216e^{-6x}$$

$$f^{(4)}(x) = (-6)^4 e^{-6x} = 1296e^{-6x}$$

$$f(-4) = e^{-6(-4)} = e^{24}$$

$$f'(-4) = -6e^{-6(-4)} = -6e^{24}$$

$$f''(-4) = 36e^{-6(-4)} = 36e^{24}$$

$$f'''(-4) = -216e^{-6(-4)} = -216e^{24}$$

$$f^{(4)}(-4) = 1296e^{-6(-4)} = 1296e^{24}$$

$$P_0(x) = f(-4) = e^{24}$$

$$P_1(x) = f(-4) + f'(-4)(x+4) = e^{24} + (-6e^{24})(x+4) \\ = e^{24} - 6e^{24}(x+4)$$

$$P_2(x) = f(-4) + f'(-4)(x+4) + \frac{f''(-4)(x+4)^2}{2!}$$

$$= e^{24} - 6e^{24}(x+4) + \frac{36e^{24}}{2!}x(x+4)^2$$

$$= e^{24} - 6e^{24}(x+4) + 18e^{24}(x+4)^2$$

$$P_3(x) = f(-4) + f'(-4)(x+4) + \frac{f''(-4)(x+4)^2}{2!} + \frac{f'''(-4)(x+4)^3}{3!}$$

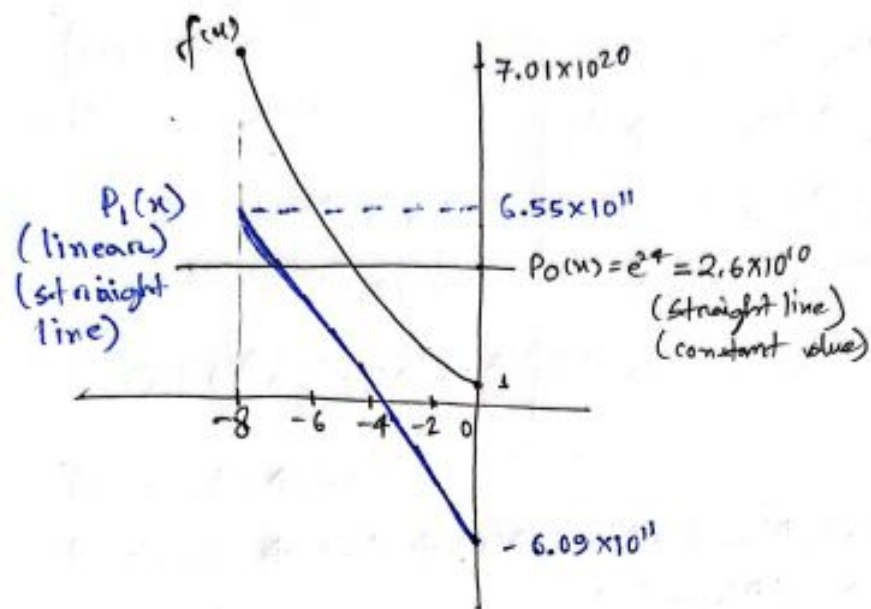
$$= e^{24} - 6e^{24}(x+4) + 18e^{24}(x+4)^2 + \frac{-216e^{24}(x+4)^3}{3!}$$

$$= e^{24} - 6e^{24}(x+4) + 18e^{24}(x+4)^2 - 36e^{24}(x+4)^3$$

$$P_4(x) = f(-4) + f'(-4)(x+4) + \frac{f''(-4)(x+4)^2}{2!} + \frac{f'''(-4)(x+4)^3}{3!} + \frac{f^{(4)}(-4)(x+4)^4}{4!}$$

$$= e^{24} - 6e^{24}(x+4) + 18e^{24}(x+4)^2 - 36e^{24}(x+4)^3 + \frac{1296e^{24}(x+4)^4}{4!}$$

$$= e^{24} - 6e^{24}(x+4) + 18e^{24}(x+4)^2 - 36e^{24}(x+4)^3 + 324e^{24}(x+4)^4$$



$$f(u) = e^{-6u}$$

$$\text{for } u > 0, f(u) \approx e^{-6 \times 0} = 1$$

$$\text{for } u = -8, f(u) = e^{-6 \times -8} = e^{48} = 7.01 \times 10^{20}$$

$$P_0(x) = e^{24} = 2.6 \times 10^{10} \text{ (constant value)}$$

$$P_1(x) = e^{24} - 6e^{24}(x+4)$$

$$= -6xe^{24} + e^{24} - 24e^{24}$$

$$\text{linear equation} \rightarrow = -1.58 \times 10^{11}x - 6.09 \times 10^{11}$$

(straight line with a negative gradient of -1.58×10^{11})

$$\text{Put } x = 0, P_1(x) = -6.09 \times 10^{11}$$

$$x = -2, P_1(x) = -2.93 \times 10^{11}$$

$$x = -8, P_1(x) = 6.55 \times 10^{11}$$

It is not necessary to draw $P_1(x)$. Only drawing $P_0(x)$ is enough for the question